

Functional UAT Checklist – Drought & Crop Risk Monitor

This Functional User Acceptance Testing (UAT) document validates the behavior, data integrity, visualization logic, and intelligence-dashboard interactions of the Drought & Crop Risk Monitor platform. The application is designed to monitor global drought conditions, agricultural stress exposure, and crop-risk intelligence using a modern, interactive analytics dashboard powered by FastAPI, Next.js, and Leaflet-based geospatial visualization.

Phase 1: Input Package Validation

Validation Area	Expected Result	Status
Backend API	FastAPI server responds successfully on localhost:8000	PASS
Frontend Dashboard	Next.js dashboard renders successfully on localhost:3000	PASS
Mock Data	Risk dataset loads correctly from mock_data.json	PASS
Map Rendering	Leaflet map loads successfully with risk markers	PASS

Phase 2: Visual Audit Review (VAR)

The dashboard must follow the intended intelligence-style interface with dark-theme styling, split-screen visualization, responsive behavior, interactive controls, collapsible panels, and animated UI interactions.

Audit Area	Expected Result	Status
Layout Verification	Dashboard maintains map + sidebar structure	PASS
Dark Theme Styling	Dark intelligence theme applied consistently	PASS
Responsive Design	UI adapts properly across desktop views	PASS
Micro Interactions	Hover effects and transitions work smoothly	PASS
Collapsible Panels	Expandable sections function correctly	PASS

Phase 3: Functional UAT Test Cases

Test ID	Category	Test Case	Expected Result	Status
TC_001	API Connectivity	Load dashboard	Frontend fetches backend data successfully	PASS
TC_002	Map Rendering	Open map view	Leaflet map renders without errors	PASS
TC_003	Marker Display	Load risk markers	Markers appear in correct geographic positions	PASS
TC_004	Risk Classification	Verify risk labels	High/Medium/Low risk displays correctly	PASS

TC_005	Sidebar Metrics	View analytics cards	Metrics cards display accurate values	PASS
TC_006	Filter Controls	Click filter buttons	Map updates based on selected filter	PASS
TC_007	Search Feature	Search country name	Matching regions appear dynamically	PASS
TC_008	Download Feature	Click Download Data	Data file downloads successfully	PASS
TC_009	Popup Information	Click map marker	Popup shows region and risk details	PASS
TC_010	Collapsible Panels	Expand/collapse cards	Panels open and close correctly	PASS
TC_011	Hover Animation	Hover over buttons/cards	Glow and transitions activate smoothly	PASS
TC_012	Data Integrity	Validate API values	Risk scores match backend calculations	PASS
TC_013	Coordinates Validation	Verify coordinates	Latitude and longitude render correctly	PASS
TC_014	Responsive Behavior	Resize browser	Dashboard remains visually stable	PASS
TC_015	Error Handling	Disconnect backend API	Frontend handles fetch failure gracefully	PASS

Final Validation Summary

The Drought & Crop Risk Monitor successfully satisfies the intended functional, visual, and interaction requirements defined during the User Acceptance Testing process. The platform demonstrates stable backend integration, real-time dashboard behavior, interactive geospatial visualization, intelligence-style UI consistency, and scalable monitoring capabilities suitable for advanced drought-risk analytics systems.